Week 5: Modern Physics SOLUTIONS

Discussion Questions

1a This only occurs if the zinc plate is negatively charged. The UV light photons have enough energy to release electrons from the surface (if it is clean). They are then repelled away from the plate.

1b Electrons are released, but the Zn plate is now positively charged, and they are attracted back to the plate

2 electrons are released from the metal plate and move across to the electrode; this current completes the circuit; the battery provides a stopping voltage to prevent current below certain kinetic energies.

3 One is due to elastic scattering of the x-ray off the target nucleus (0.0709 nm), the other is due to inelastic scattering off outer/loosely bound electrons (0.731/0.0749 nm). Both are considered Compton scattering – the same equation is used to calculate both, but the mass term is changed to be either the electron or the nucleus. However, the elastic scattering of the x-rays from the nucleus also gets the label "Thomson scattering", which is the low energy limit case for Compton scattering, where the scattered x-rays energy and frequency essentially do not change.

4 (b) A potential barrier keeps the electrons bound, so if enough energy put in (by photon) they will be freed. $E = hf$ for incident photons.

5 (b) The Compton equation for wavelength shift was derived from conservation of energy and momentum.

6 Compton shift is still there but is *very* small (pm) compared to the wavelength of light (hundreds of nm) and hence difficult to detect. In contrast, small wavelength x-rays receive higher proportional shift during Compton scattering.

7 As $p = \sqrt{2mE}$ & $\lambda = \frac{h}{p}$, the proton will have the greater momentum, thus the smaller wavelength

8 Bohr's ad hoc (quantum) postulates –

- Electrons are only allowed to travel in certain orbits. They do not radiate (by fiat) while in their orbits.
- Electrons may discontinuously charge orbits – quantum jump. When this happens a single photon of light is emitted of frequency $f=\Delta E/h$, where ΔE is the energy difference between the orbits.
- The permitted orbits are characterized by quantized values of the orbital angular momentum:

[Remaining postulate is that Classical Mechanics applies provided the electron stays in the same orbit.]

9 See textbook HRW 11th Ed., §39.4, or Wikipedia: "*the behavior of systems described by the theory of quantum mechanics (or by the old quantum theory) reproduces classical physics in the limit of large quantum numbers. In other words, it says that for large orbits and for large energies, quantum calculations must agree with classical calculations.*" Note somewhat different to formal limits, as the equations produce the same result without simplifying to each other.

Problem Solving Questions

10a Sum of 1.13 and 4.24 = 5.37 eV = 8.59E-19 J

b Text (HRW 10/e) *p*1155 "the least energy required by an electron to escape the electric forces holding electrons in the metal"

c 0eV (from electrons below the surface, only just enough energy to escape)

d No, this would reduce the photon energy to 2.685 eV, which is less than the work function

e Use $E = hc/\lambda \rightarrow \lambda = \frac{6.63\times10^{-34}\text{Js}\times3\times10^{8}\text{ms}^{-1}}{(1.13\text{ eV} + 4.24\text{ eV})\times1.6\times10^{-19}\text{J/eV}} = \mathbf{231\ nm}$

f It would not change; KE of electrons is determined by the incident photon frequency

11a Use $E = hc/\lambda \rightarrow \lambda = \frac{6.63\times10^{-34}\text{Js}\times3\times10^{8}\text{ms}^{-1}}{(40\times10^{3}\text{ eV})\times1.6\times10^{-19}\text{J/eV}} = \mathbf{31\ pm}$

b Use $\Delta\lambda = \frac{2h}{mc}$; occurs at 180° scattering $\rightarrow \Delta\lambda = \frac{2\times6.63\times10^{-34}\text{Js}}{9.11\times10^{-31}\text{kg}\times3\times10^{8}\text{ms}^{-1}} =$ $\mathbf{4.86\times10^{-12}m}$.

8c The Compton shift is inversely proportional to the mass of the scattering object, as atoms are thousands of times more massive than free electrons, so the shift will be thousands of time less. Hence scattering off whole atoms and is virtually elastic – i.e. Thomson scattering.

12 Use $P = nhc/\lambda$, where n is the number emitted per second.

13 First, find an *approximate* value of e by taking differences between all the given data. Now assign integers n using this approximate value of e (4.1e is clearly 4e, 5.13e is 5e, and so on). Now calculate an e value for each of these, and average. You should get: 1.641×10^{-19} C.

14i Effective cross-section of planet from the star is πr^2; so energy in one second = $\pi r^2 \times 1500\text{W/m}^2 \times 1\text{s}$

ii Use $P = \varepsilon\sigma AT^4 = 1.93 \times 10^{17}$ W (note that as a perfect black body, emissivity $\varepsilon = 1$, and A here is <u>total</u> surface $4\pi r^2$) to give T = 285 K = **12°C**.

iii Use Wien's Law $\lambda_{MAX} = 2.98777 \times 10^{-3}/T$; answer 1.0×10^{-5} m.

15a insert quantities in SI units (carefully) to get 13.6 eV. Most scientific calculators cannot handle the number of floating points required if the values are entered without care to minimize the powers of the constants first, e.g. the numerator may be to 71st power, the denominator greater than 100th!

b Use $E = hf = 13.6(\frac{1}{m^2} - \frac{1}{n^2})$eV with $m = 3$ and $n = 2 \rightarrow E = 13.6eV\left(\frac{1}{3^2} - \frac{1}{2^2}\right) = 1.89$ eV

$\rightarrow \lambda = \frac{hc}{E} = \frac{4.15\times10^{-15}\text{eVs}\times3\times10^{8}ms^{-1}}{1.89\text{ eV}} = \mathbf{6.56\ nm}$

c The equivalent formula for 'hydrogen-like' atoms uses $E = 13.6\left(\frac{1}{m^2} - \frac{1}{n^2}\right)Z^2$, for He^+ $Z = 2$; lowest energy transition is $m = 3$ to $n = 2$. $\mathbf{E = 7.6\ eV}$ is comfortably into the ultraviolet (λ=163 nm).

16a the equality of the pattern spacing means that the wavelengths are ~ the same, since this is a wave phenomenon **b** $p = h/\lambda = \frac{6.63\times10^{-34}\text{Js}}{30\times10^{-12}\text{m}} = 2.21\times10^{-23}\text{kgms}^{-1}$

c $E = p^2/2m = \frac{h^2}{2m_e\lambda^2} = 2.7\times10^{-16}$ J = 1.7 keV

17a see text HRW Principles of Physics, Section 36.7, pp1104-5 on X-Ray diffraction.

b insert $n = 1$ and $n = 2$ into the Bragg formula ($2d\sin\theta = n\lambda$)

$$\lambda_{n=1} = 2\times0.3\times10^{-9}\text{m}\times\sin 30° \div 1 = 3.0\times10^{-10}\text{ m}$$
$$\lambda_{n=2} = 2\times0.3\times10^{-9}\text{m}\times\sin 30° \div 2 = 1.5\times10^{-10}\text{ m}$$

18 Full details on the operation of x-ray generation via electron beams onto a spinning tungsten disc can be found on reputable websites. Two kinds of x-rays are produced as the beam of electrons interacts with the tungsten target disc – characteristic and continuous/ bremsstrahlung radiation. The former are when the incident electron knocks out a shell electron from the target, causing a higher electron to quantum jump to the lower energy state, releasing the difference as a x-ray photon. Continuous spectra or bremsstrahlung radiation x-rays are when the electron is slowed by the interaction with the target, and the difference in kinetic energy is released as a photon. These can carry up to as much energy as the incident electron initially carried. This creates a lower wavelength limit, generally at a lower value than any characteristic x-rays.

→Maximum KE of the x-rays will be 40 keV, set this equal to hc/λ; answer is $\lambda_{min} = 31$ pm.

19 insert values carefully - $\alpha \approx \frac{1}{137.035...}$ and it's unit-less!

Past Exam Questions

20 Use the Compton shift formula: $\Delta\lambda = \frac{h}{mc}(1 - \cos\theta)$; since m is the mass of the electron, the answer does not change for part (b)

21 (a) Use $E = hc/\lambda$ (b) Use $P = nE/t = nhc/\lambda t$, and solve for n/t, where n = number of photons

22a This is Bragg scattering (or 'reflection'), see text section 36.7; electrons acting *wave-like*. **b** KE of electrons is 55 eV; $p = \sqrt{2mE} = \sqrt{2\times9.1\times10^{-31}\text{kg}\times88\times10^{-19}\text{J}} = 4\times10^{-24}\text{kg.ms}^{-1}$ **c** Use the Bragg scattering formula ($2d\sin\theta = n\lambda$) → $d = \frac{n\lambda}{2\sin\theta} = \frac{nh}{2p\sin\theta} = 9.14\times10^{-11}$ m

23ai The Photoelectric effect, whereupon a photon of the correct frequency/wavelength when incident upon the metal, carries the correct energy to eject an electron. If the laser's wavelength produces photons which exactly matches the work function of the metal, the electrons are ejected with no kinetic energy.

23aii Use $eV = 0.5mv^2 \rightarrow v = \sqrt{\frac{2eV}{m}} = 1.45\times10^7 ms^{-1} (\approx 0.05c)$

23aiii $\lambda = \frac{h}{p} = \frac{h}{mv} = 5.0\times10^{-11}$m

23b Diffraction (spreading of the wave-front/beam) is inversely proportional to aperture size [see Rayleigh criteria for limits on identification of two sources – as circular aperture size decreases, angular separation required for identification of resultant diffraction maxima increases]. This will not produce the desired result – the spot will now *spread.*

24a see text section p1172ff – "The probabilistic nature of quantum physics places an important limitation on detecting a particle's position and momentum. That is, it is not possible to measure the position and the momentum of a particle simultaneously with unlimited precision."

24b $p = \sqrt{2m_eE} = 1.99\times10^{-24}\text{kg.ms}^{-1}$

24c $\Delta x\Delta p > h/4\pi \rightarrow \Delta x \geq \frac{\hbar}{2\Delta p} \sim \frac{\hbar}{2p} \sim 10^{-11} - 10^{-10}$ metres (cf. Bohr radius 5.2×10^{-11} m)

25a Photon frequency $f = \frac{c}{\lambda} = 5\times10^{14}$ Hz (cf. photon energy $E = \frac{hc}{\lambda} = 3.31\times10^{19}$ J)

Photon frequency = frequency of electron orbit

Centripetal force $F = \frac{m_ev^2}{r} \equiv \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2}$ = Coulomb force

Electron velocity $v = \sqrt{\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{m_er}}$

Electron orbit frequency $f = \frac{1}{T} = \frac{v}{2\pi r} = \sqrt{\frac{1}{16\pi^3\varepsilon_0}\frac{Ze^2}{m_er^3}}$

Electron radius $r = \left(\frac{Ze^2}{16\pi^3\varepsilon_0 m_e f^2}\right)^{\frac{1}{3}}$

Assume hydrogen $Z = 1$

$r = \left(\frac{(1.6\times10^{-19})^2}{16\pi^3\times8.85\times10^{-12}\times9.11\times10^{-31}\times(5\times10^{14})^2}\right)^{\frac{1}{3}} = 2.95\times10^{-10}$ m = **2.95 Å**

25b $\Delta p / p = 10^{-6}$ implies $\Delta p = 10^{-6} p$

Uncertainty principle: $\Delta x \Delta p = h / 4\pi$ implies $\Delta x = \frac{h}{4\pi\Delta p}$

and we have $p = \frac{h}{\lambda}$ for the photon, so $\Delta x = \frac{h}{4\pi(10^{-6}p)} = \frac{\lambda}{4\pi10^{-6}}$

With λ = 600 nm, we get Δx = **0.047 m**.

25c $E_n = \frac{h^2}{8m_eL^2}n^2 = \frac{(6.63\times10^{-34}\text{ Js})^2}{8\times9.11\times10^{-31}\text{ kg}\times(0.2\times10^{-9}\text{m})^2} = 1.51\times10^{-18}n^2$ (Joules)

$$\therefore E_{n=1} = 1.51\times10^{-18}\text{ J} = 9.42\text{ eV}$$
$$E_{n=2} = 6.04\times10^{-18}\text{ J} = 37.7\text{ eV}$$
$$E_{n=3} = 1.36\times10^{-17}\text{ J} = 84.8\text{ eV}$$